Number sets and intervals

Sapienza Università di Roma

Outline

The five main number sets
- The natural numbers and the integers
- The rational numbers
- The real numbers
- The complex numbers

Intervals

The five main number sets

Introduction: The five main number sets

- In this section we recall the five fundamental number sets: $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$
- We will see how rational numbers and real numbers are written in decimal form
- We will understand that there exist real numbers that are not rational and that the number sets are contained in one another
- These sets are the "ground" on which all of Mathematical Analysis is built

The natural numbers

$\mathbb{N}$ is the set of **natural numbers**, that is, the numbers used for counting:

$$\mathbb{N} = \{0,\ 1,\ 2,\ 3,\ 4,\ \dots\}$$

- We recall the convention adopted in these notes: **zero is a natural number**, that is $0 \in \mathbb{N}$
- We denote by $\mathbb{N}_{>0}$ the set of **positive natural numbers**:

$$\mathbb{N}_{>0} = \mathbb{N} \setminus \{0\} = \{1,\ 2,\ 3,\ 4,\ \dots\}$$

What we already know about the natural numbers

In the chapter *Sets* we have already introduced, for the natural numbers:

- the definitions of **even** number ($n = 2\,m$) and **odd** number ($n = 2\,m + 1$), with $m \in \mathbb{N}$, and the fact that every natural number is either even or odd, but never both
- the **well-ordering axiom**, stating that every non-empty subset of $\mathbb{N}$ has a minimum

The integers

$\mathbb{Z}$ is the set of **integers** (also called signed numbers), that is:

$$0,\ 1,\ -1,\ 2,\ -2,\ \ldots$$

The rational numbers

$\mathbb{Q}$ is the set of **rational numbers**, that is, of the fractions:

$$\frac{n}{m}, \quad \text{where} \quad n, m \in \mathbb{Z}, \quad \text{and} \quad m \neq 0.$$

Rational numbers can also be written in decimal form. For example:

$$\frac{3}{4} = 0.75, \quad \text{while} \quad \frac{1}{3} = 0.333\cdots = 0.\overline{3}.$$

A rational number, written in decimal form, has a **finite or infinite periodic decimal expansion**: after the decimal point there is a finite number of digits, or an infinite number of digits which, from a certain point on, repeat *periodically*.

The number $0.\overline{9}$

The same rational number can have two different decimal expansions. The classical example is the equality

$$0.\overline{9} = 1$$

for which we gave four different proofs in the chapter *Sets*, and which we therefore do not repeat here.

The real numbers

$\mathbb{R}$ is the set of **real numbers**, that is, those which are identified with **finite or infinite decimal expansions, periodic or non-periodic**: after the decimal point there may be any string of digits, possibly *infinite* and *non-periodic*.

That numbers of this last kind exist (that is, real but not rational), can be understood by reflecting on examples such as:

$$0.10110111011110\ldots$$

After the decimal point, the previous number has: one digit equal to 1, then 0, then two digits equal to 1, then 0, then three digits equal to 1 . . . and so on.

It is clear that this rule defines a decimal number precisely. On the other hand, the string of digits after the decimal point is neither finite nor periodic: therefore this number is irrational.

The complex numbers

$\mathbb{C}$ is the set of **complex numbers**, that is, numbers of the form

$$a + i\, b,$$

where a, b are real numbers, and i is the *imaginary unit*, that is, a number whose square is -1.

Inclusions among the number sets

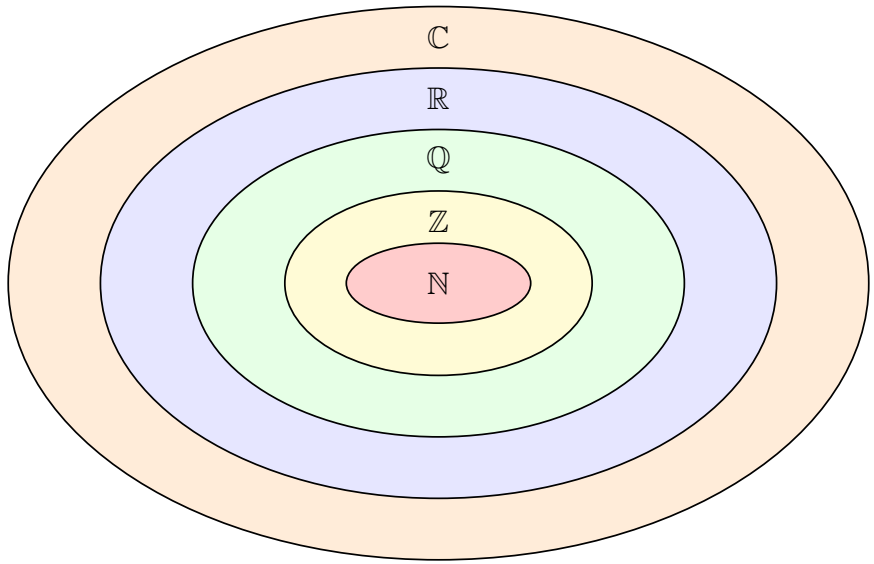
Among the number sets introduced, the following inclusions hold:

$$\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C}.$$

As indicated by the symbols, all the inclusions are strict:

- there exist integers that are not natural numbers (the negative numbers),
- there exist rational numbers that are not integers (the proper fractions),
- there exist real numbers that are not rational (the irrational numbers),
- there exist complex numbers that are not real (the imaginary numbers).

The number sets, one inside the other



Positive and non-negative numbers

As for $\mathbb{N}_{>0}$, we denote by

$$\mathbb{Z}_{>0}, \qquad \mathbb{Q}_{>0}, \qquad \mathbb{R}_{>0}$$

the sets of **positive** integer, rational and real numbers, and by

$$\mathbb{Z}_{\geq 0}, \qquad \mathbb{Q}_{\geq 0}, \qquad \mathbb{R}_{\geq 0}$$

the sets of **non-negative** integer, rational and real numbers (that is, positive or zero).

For example:

$$\mathbb{R}_{>0} = \{x \in \mathbb{R} : x > 0\} \qquad \text{and} \qquad \mathbb{Z}_{\geq 0} = \{x \in \mathbb{Z} : x \geq 0\} = \mathbb{N}$$

Summary

- $\mathbb{N}$: natural numbers, with $0 \in \mathbb{N}$ and $\mathbb{N}_{>0} = \mathbb{N} \setminus \{0\}$; $\mathbb{Z}$: integers
- $\mathbb{Q}$: fractions $\frac{n}{m}$ with $n, m \in \mathbb{Z}$, $m \neq 0$, that is, finite or infinite periodic decimal expansions
- $\mathbb{R}$: arbitrary decimal expansions, even infinite and non-periodic (irrational numbers exist)
- $\mathbb{C}$: numbers $a + i\,b$ with $i^2 = -1$
- $\mathbb{N} \subsetneqq \mathbb{Z} \subsetneqq \mathbb{Q} \subsetneqq \mathbb{R} \subsetneqq \mathbb{C}$
- Notations $\mathbb{Z}_{>0}$, $\mathbb{Q}_{>0}$, $\mathbb{R}_{>0}$ and $\mathbb{Z}_{\geq 0}$, $\mathbb{Q}_{\geq 0}$, $\mathbb{R}_{\geq 0}$

Intervals

Introduction: Intervals

- In this section we define **intervals**, the most used subsets of $\mathbb{R}$ in Analysis
- We will distinguish closed, open, bounded and unbounded intervals
- We will see the **connectedness** property that characterizes them and the Cartesian product of intervals
- Intervals will be the natural domains of the functions we will study

Bounded intervals

Definition 1: of bounded interval

Given two real numbers a, b, a ***bounded interval*** *with endpoints a and b is one of the following sets:*

$$[a,b] = \{x \in \mathbb{R} : a \leq x \leq b\}, \quad [a,b) = \{x \in \mathbb{R} : a \leq x < b\}$$

$$(a,b] = \{x \in \mathbb{R} : a < x \leq b\}, \quad (a,b) = \{x \in \mathbb{R} : a < x < b\}$$

Closed and open intervals

- The square (round) bracket at one of the two endpoints indicates that this endpoint is included in (excluded from) the interval.
- The intervals $[a, b]$ are called **closed**; the intervals (a, b) are called **open**.

Unbounded intervals

Definition 2: of unbounded interval

Given a real number a, an ***unbounded interval*** *(or* ***half-line****) with endpoint a is one of the following sets:*

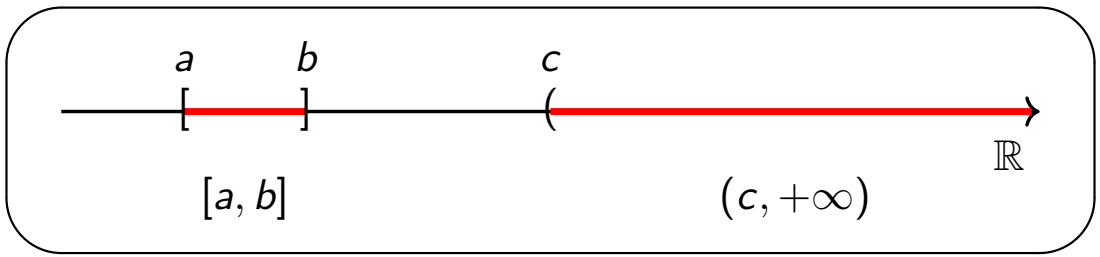
$$[a, +\infty) = \{x \in \mathbb{R} : x \geq a\}, \quad (a, +\infty) = \{x \in \mathbb{R} : x > a\}$$

$$(-\infty, a] = \{x \in \mathbb{R} : x \leq a\}, \quad (-\infty, a) = \{x \in \mathbb{R} : x < a\}$$

Unbounded intervals

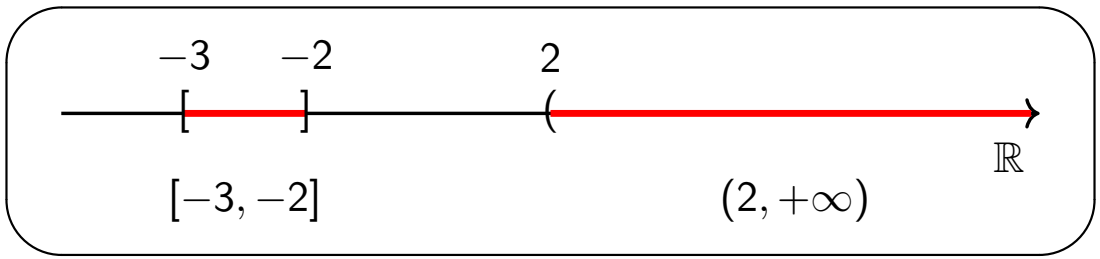
- The symbols $-\infty$ and $+\infty$ **are not real numbers**: they are not endpoints that can belong to the set, and therefore next to them one always writes a round bracket.
- The whole line is an unbounded interval as well:

$$\mathbb{R} = (-\infty, +\infty)$$



Example

Intervals. The graphical representation of the closed bounded interval $[-3, -2]$ and of the open unbounded interval $(2, +\infty)$ is the following:



The first one contains both of its endpoints, the second one does not contain its endpoint 2 and is not bounded above.

Connectedness

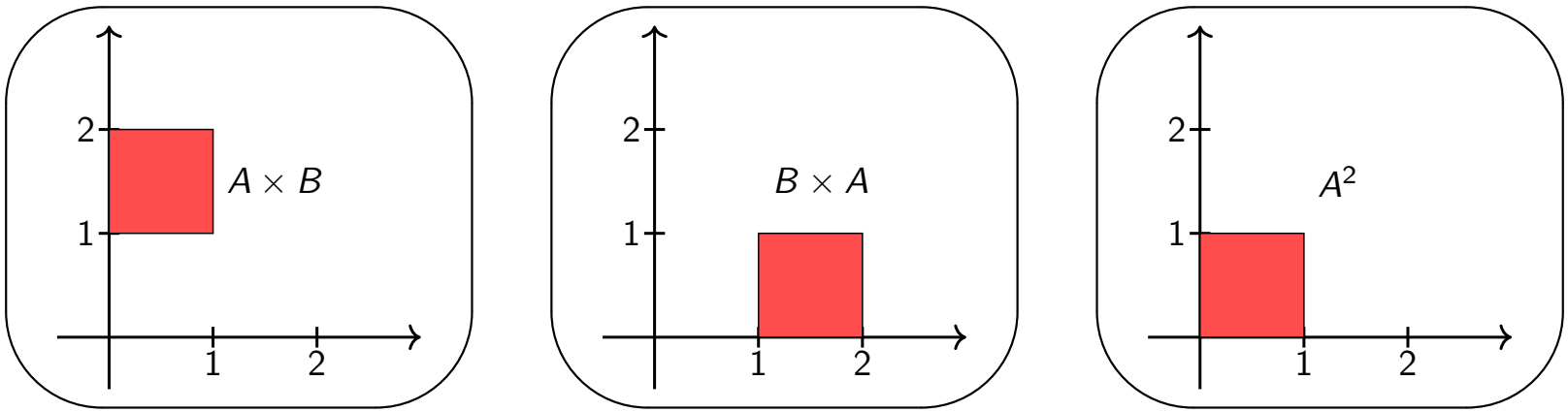
It can be proved that the intervals, bounded or unbounded, are exactly the subsets I of $\mathbb{R}$ that satisfy the following property (called **connectedness**):

$$\forall x_1, x_2, x_3 \in \mathbb{R} \text{ such that } x_1 < x_2 < x_3, \text{ if } x_1, x_3 \in I, \text{ then } x_2 \in I$$

- In what follows we will sometimes consider the Cartesian product of two (or more) intervals, which can be given the geometric meaning of a rectangle (in two dimensions) or of a parallelepiped (in three dimensions).

Example

Cartesian product of intervals. Let $A = [0, 1]$, $B = [1, 2]$; the figure



illustrates the sets $A \times B$, $B \times A$, $A \times A$, also denoted by A^2. In general, $A \times B$ is different from $B \times A$.

Summary

- Bounded intervals with endpoints a, b: $[a, b]$, $[a, b)$, $(a, b]$, (a, b)
- Square bracket: endpoint included; round bracket: endpoint excluded
- $[a, b]$ **closed**, (a, b) **open**; the half-lines $[a, +\infty)$, $(a, +\infty)$, $(-\infty, a]$, $(-\infty, a)$ and $\mathbb{R} = (-\infty, +\infty)$ are the unbounded intervals
- Intervals are exactly the subsets of $\mathbb{R}$ with the **connectedness** property
- The Cartesian product of intervals is a rectangle; in general $A \times B \neq B \times A$

Summary

Number sets:

- $\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C}$, all strict inclusions
- Rational numbers have finite or periodic decimal expansion (for example $0.\overline{9} = 1$)
- Real numbers admit infinite non-periodic decimal expansions (irrational numbers)

Intervals:

- Closed, open, half-open, bounded and unbounded
- Characterized by the connectedness property
- Their Cartesian product has a geometric meaning (rectangles, parallelepipeds)